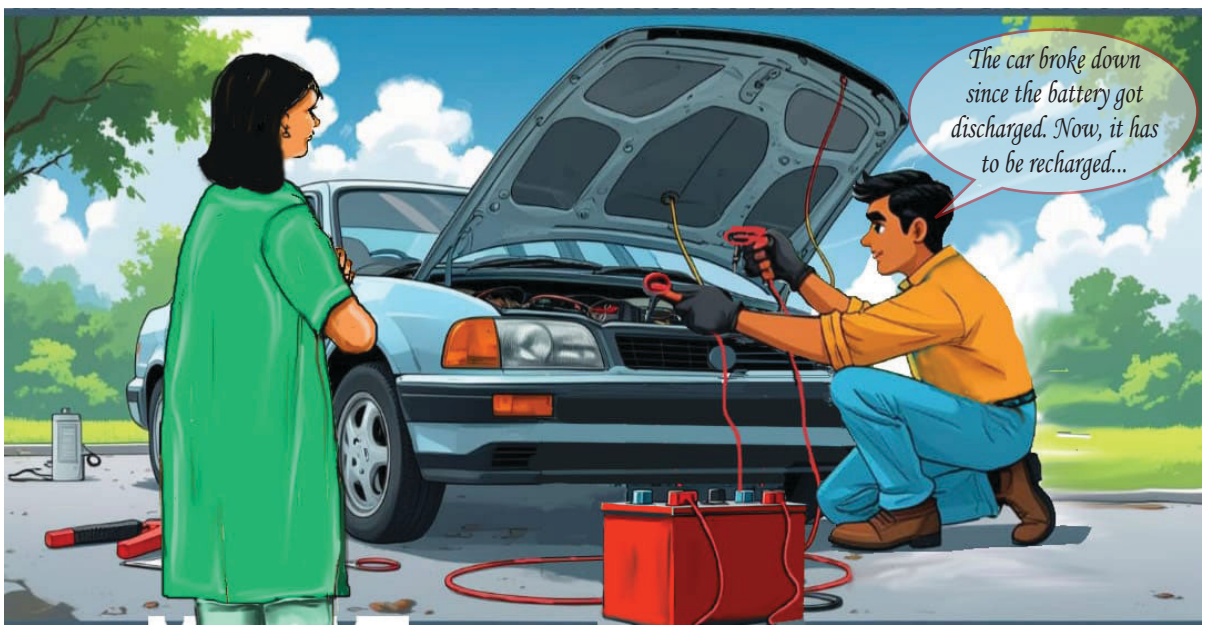
5 ELECTROCHEMISTRY



We usually travel in vehicles. You can imagine how difficult it would be if vehicles breakdown midway due to electrical problems. Similar circumstances arise when mobile phones and laptops cannot be operated. We cannot even think of such situations these days. To operate all these, stored electrical energy is required. Cells or batteries are required as power sources to operate devices such as inverters, emergency lamps, watches, clocks, various remote controls, portable music devices, headphones, hearing aids, medical equipments for diagnosis and treatment, electrically operated toys etc.

Have you ever thought about how a cell stores electrical energy and makes it available as required? This is made possible by electrochemical reactions.

Electrochemistry is the branch of chemistry which deals with the studies on the processes that produce electricity through chemical reactions and use electricity to bring about chemical reactions. Electrochemical cells are the devices that make such changes possible.

Electrochemical cells can be divided into two types.

1. Galvanic cells
2. Electrolytic cells

Chemistry and Electricity

In 1794, Alessandro Volta discovered that electricity can be produced by placing two different metals (Zn,Cu) on either side of a piece of moist paper. In 1800, Voltas battery was used as a source of electricity to prove that water can be split into oxygen and hydrogen, and it was one of the most important experiments in the history of chemistry. This experiment indicated that hydrogen and oxygen atoms are associated with positive and negative electrical charges, and that this could be the reason for the bond between them. At the same time, the use of electricity as a means of effecting chemical reactions also played an important role in the development of chemistry. Humphry Davy was the first to produce elemental sodium by the electrolysis of molten sodium hydroxide. Michael Faraday was able to prove that there is a relationship between the amount of electric charge flowing through a solution and the amount of products of electrolysis. But we had to wait until the end of the nineteenth century to discover the electron.

Galvanic cells

Galvanic cells are devices that convert chemical energy into electrical energy. Daniel cell is an example of a galvanic cell. The method of constructing a Daniel cell is given below. Let us do this activity.

Materials required

- Beaker
- Zinc rod
- Copper rod
- Zinc sulphate solution
- Copper sulphate solution
- Salt bridge
- NH_4Cl / KCl / KNO_3
- Copper wires
- Galvanometer
- Voltmeter
- LED bulb

Experimental procedure

Dip a zinc rod in the zinc sulphate solution taken in a beaker as shown in Figure 5.1 (Half cell 1). Dip a copper rod in the copper sulphate solution in another beaker (Half cell 2). Connect the solutions using a salt bridge. (Take a piece of rolled filter paper and add a little potassium chloride or ammonium chloride or potassium nitrate into it and then sprinkle water to moisten it. This filter paper can be folded into 'U' shape and immersed in the two beakers connecting the two solutions in them and it can be used as a salt bridge.)

Connect the zinc rod and copper rod to the bulb using copper wires.

What did you observe?

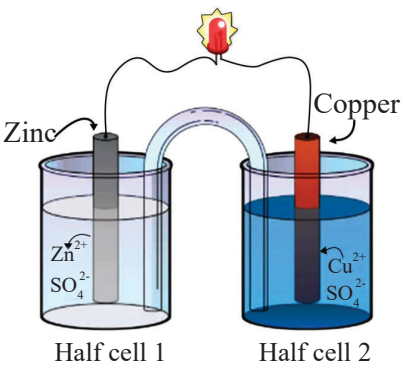


Fig. 5.1

What is the reason for the glowing of the bulb?

We can understand that there is a flow of electric current through the bulb.

Separate the copper wires from the bulb and connect them to a galvanometer as shown in Figure 5.2. When the copper rod is connected to the positive terminal of the galvanometer and the zinc rod to the negative terminal, it can be seen that the pointer of the galvanometer is deflecting towards the right (towards the positive terminal).

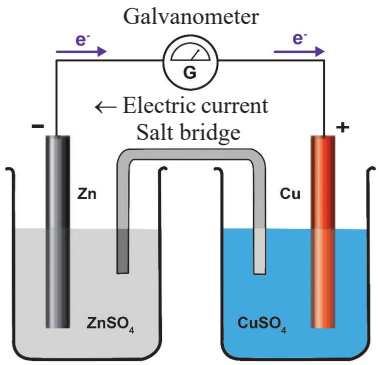


Fig. 5.2

From this, it is clear that the flow of electrons is from the zinc rod to the copper rod and that of electric current is in the opposite direction.

Now, connect a voltmeter instead of the galvanometer as shown in Figure 5.3 and find the voltage between the two electrodes.

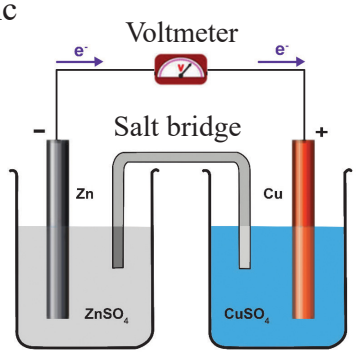


Fig. 5.3

The reaction that occurs at each electrode is given below.

Zinc electrode : $Zn(s) \rightarrow Zn^{2+}(aq) + 2e^-$

What type of chemical reaction takes place at the zinc electrode? (Oxidation / Reduction) .....

Copper electrode : $Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$

What type of chemical reaction takes place at the copper electrode?

It can be understood that oxidation occurred at the zinc electrode and reduction at the copper electrode.

The electrode where oxidation occurs is called the anode, and the electrode where reduction occurs is called the cathode.

The processes that takes place in the Daniel cell can be summarised as:

Anode : $Zn(s) \rightarrow Zn^{2+}(aq) + 2e^-$

Cathode : $Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$

Cell reaction :

$$Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$$

Salt bridge

A salt bridge is a U-shaped glass tube filled with inert electrolyte. The commonly used electrolytes are potassium chloride (KCl), ammonium chloride (NH_4Cl) and potassium nitrate (KNO_3) which are made into a gel using agar-agar. Salt bridge is used to connect the oxidation and reduction half cells of a galvanic cell. It helps to establish an electrical contact between the two solutions in such a way that they do not mix with each other. It is the salt bridge that enables smooth cell operation by ensuring electrical neutrality within the internal circuit.

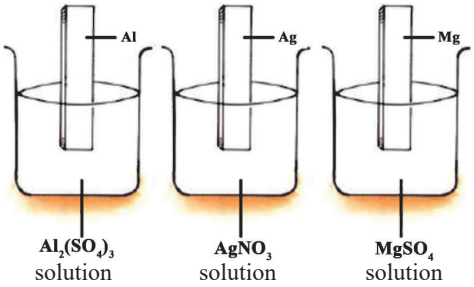


Fig. 5.4

The electric current is produced as a result of a redox reaction taking place in the galvanic cell.

Examine the metal rods and the solutions in which they are immersed in Figure 5.4.

In the Daniel cell, replace the zinc rod in zinc sulphate by aluminium, silver and magnesium rods (plates) and immerse these metals in their own salt solution. Repeat the experiment by connecting each to the copper rod immersed in copper sulphate.

List out the direction of flow of electrons, equation of reaction at each electrode, cell voltage and analyse them.

Electrode pair	Direction of flow of electrons	Anode	Cathode	Anode reaction	Cathode reaction	Cell reaction	Cell voltage
Al - Cu							
Ag -Cu							
Mg - Cu							

Table 5.1

- Which has the higher value of voltage, Zn - Cu galvanic cell or Al - Cu galvanic cell? .....

  It is clear that the tendency of aluminium to get oxidised in redox reactions is greater than that of zinc. That is, aluminium is more reactive than zinc.
- Which has the higher value of voltage, Mg - Cu galvanic cell or Al - Cu galvanic cell?

  It can be found that the value of voltage for the Mg - Cu galvanic cell is higher.
- Here, what is the direction of flow of electrons in a galvanic cell made up of a pair of copper and silver electrodes?

  .....

  It is found that the direction of flow of electrons is from the copper electrode to the silver electrode.

? Draw a diagram of the galvanic cell having silver and copper as electrodes and mark the direction of flow of electrons.

Prepare a list of the metals, familiarised through the experiment, in the order of their tendency to get oxidised in redox reactions.

Metal	
Mg	Oxidation tendency (reactivity) decreases ↓
.....	
.....	
.....	
Ag	

Table 5.2

In this way, we can construct galvanic cells using different elements as electrodes and compare their reactivities. The series in which elements are arranged in the decreasing order of their reactivity is known as reactivity series. Examine the reactivity series given below.

Potassium	K	Reactivity decreases ⟶
Sodium	Na	
Calcium	Ca	
Magnesium	Mg	
Aluminium	Al	
Zinc	Zn	
Iron	Fe	
Nickel	Ni	
Tin	Sn	
Lead	Pb	
Hydrogen	H	
Copper	Cu	
Mercury	Hg	
Silver	Ag	
Gold	Au	

Table 5.3

Applications of the reactivity series

1. To identify which metal is being displaced.

You have studied about displacement reactions in ninth standard. Let us do the experiment given below.

Place a zinc rod in a beaker of copper sulphate solution. Observe for some time.

Electrical neutrality

The formation of excess positive or negative charge in a substance leads to instability. For example, immerse a zinc rod in zinc sulphate solution. Some Zn atoms enter the solution as Zn^{2+} ions, and the electrons they release remain in the metal.

$$Zn(s) \rightarrow Zn^{2+}(aq) + 2e^-$$

As this process continues, the electrons remaining in the zinc create a negative charge on the metal, making subsequent oxidation difficult. The increase in positive charge that occurs in the aqueous solution also increases this obstruction. So the process stops very quickly. If the electrons can be removed from the metal as the positive ions go into the solution, stoppage of the process can be avoided. One method to regulate this is to bring the extra electrons outside through an external circuit. That is, to connect it with another metal that can accept electrons. For example, it can be connected to a copper rod that is immersed in copper sulphate. This system can be converted into an electrochemical cell by using a salt bridge which ensures continuity of electrical flow.

Did the colour of $CuSO_4$ solution change?

What change was observed in the zinc rod?

What may be the reason for the fading of colour of the solution?

With the help of the reactivity series, find out which metal is more reactive, Zn or Cu?

What are the ions formed when $CuSO_4$ dissolves in water?

A decrease in the concentration of Cu^{2+} ions is the reason for the fading of blue colour.

What changes have occurred to the Cu^{2+} ions?

Write the chemical equation of the reaction.

What changes have occurred to Zn?

Write the chemical equation of the reaction.

Examine the equation of the reaction which specifies the ions involved.

$$Zn + \overset{2+}{Cu}\,\overset{2-}{SO_4} \rightarrow \overset{2+}{Zn}\,\overset{2-}{SO_4} + Cu$$

Zn underwent oxidation and Cu^{2+} underwent reduction. That is, it can be seen that a redox reaction has taken place here.

Which metal is displaced here?

Take a little silver nitrate solution in a beaker and dip a copper rod in it. Observe for some time.

Is there any change in the colour of the solution?

What change was observed in the copper rod?

Examine the equation of the chemical reaction taking place here.

$$Cu + 2AgNO_3 \rightarrow Cu(NO_3)_2 + 2Ag$$

Explain the reaction by comparing the positions of Ag and Cu in the reactivity series.

Complete this chemical equation by assigning oxidation number.

$$\overset{(...)}{Cu} + 2\overset{+1}{AgNO_3} \rightarrow \overset{(...)}{Cu(NO_3)_2} + 2\overset{0}{Ag}$$

Which ion is responsible for the change in the colour of the solution?

Which metal is oxidised here?

Complete the equation for oxidation.

$$Cu \rightarrow \ldots\ldots\ldots\ldots\ldots\ldots + 2e^-$$

Which one is reduced here?

Write the equation for reduction.……....…….....

The more reactive metal displace the less reactive metal from its salt solution.

Silver nitrate solution cannot be stored in a copper vessel. Why?

2. To identify oxidising agent and reducing agent

Consider the displacement reaction given below.

$$Mg + Cu^{2+}SO_4^{2-} \rightarrow Mg^{2+}SO_4^{2-} + Cu$$

Which metal is oxidised here?

Write the equation for oxidation. ....……....……

Which metal is reduced here?

Write the equation for reduction. …....……....……

Which reaction occurs to the more reactive metal?

Oxidation / Reduction

Which reaction occurs to the ion of the less reactive metal?

Oxidation / Reduction

The more reactive metal gets oxidised while the ion of the less reactive metal gets reduced.

What is the oxidising agent and the reducing agent in this reaction? …….....

The substance that causes oxidation is called an oxidising agent and the substance that causes reduction is called a reducing agent.

In this reaction, Cu^{2+} is the oxidising agent. Similarly, Mg is the reducing agent.

What happens to the oxidising agent in this redox reaction? ....……

What happens to the reducing agent? .....……

In a redox reaction, the oxidising agent gets reduced and the reducing agent gets oxidised.

Identify the oxidising agent and reducing agent in the reaction given below.

$$2AgNO_3 + Cu \rightarrow Cu(NO_3)_2 + 2Ag$$

Oxidising agent :

Reducing agent :

In redox reactions, the more reactive metal acts as a reducing agent and the less reactive metal acts as an oxidising agent.

3. To identify the displacement of hydrogen from acid

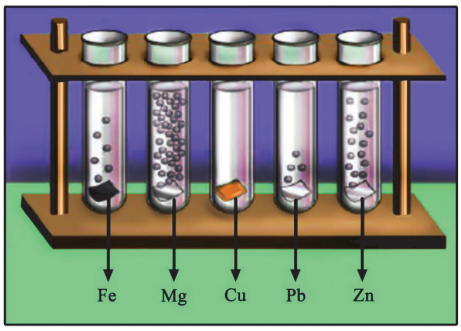


Fig. 5.5

Take equal amounts of dilute HCl in different test tubes as shown in Figure 5.5. Treat equal masses of polished Fe, Mg, Cu, Pb, Zn with dilute hydrochloric acid.

In which of the test tubes is hydrogen gas produced?

Is the rate of formation of hydrogen gas the same in all the test tubes?

Examine the position of hydrogen and the metals used here in the reactivity series and record your observations.

The metals placed above hydrogen in the reactivity series can displace hydrogen from dilute acids.

? Which of the following metals can displace hydrogen from hydrochloric acid?

Sodium, gold, silver, aluminium

Electrolytic cells

You are familiar with electrochemical cells which produce electricity through chemical changes. Likewise, there are cells which utilise electrical energy to bring about chemical changes. These cells are called electrolytic cells.

Let us do the experiment of electrolysis of acidified water.

Procedure of the experiment

Take a plastic cup. Make two holes at the bottom and attach rubber stoppers as shown in Figure 5.6. Insert carbon electrodes (graphite rods from old torch cells can be used for this purpose) into these rubber stoppers. Connect the electrodes to a 6V battery. Fill the cup with water so that the electrodes are immersed. Add 2-3 mL of dilute sulphuric acid. Dilute sulphuric acid is added to increase the ionisation of water and thereby increase its electrical conductivity. Take two test tubes filled with water and invert them over the carbon electrodes. Turn on the switch and leave the experimental setup undisturbed for some time.

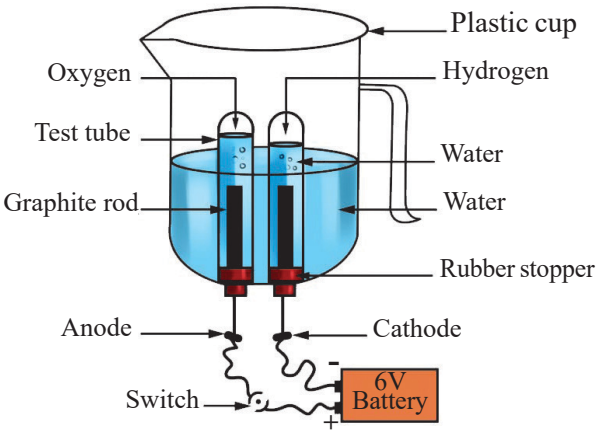


Fig. 5.6

You will observe bubbles forming at both the electrodes.

After some time, remove the test tubes carefully.

Bring a burning candle close to the mouth of the test tubes and record the observations.

.....

Which gas is produced in each test tube?

Write the chemical equation of the reaction. .....

Here, acidified water decomposes into hydrogen and oxygen. This is because of the flow of electricity from the external electrical source into the solution through the electrodes.

Electrolytes are substances that undergo chemical changes when electricity passes through them.

> Electrolytes are substances that conduct electricity in molten state or in an aqueous solution and undergo chemical changes. Electrolysis is the process by which an electrolyte undergoes chemical changes when electricity is passed through it.

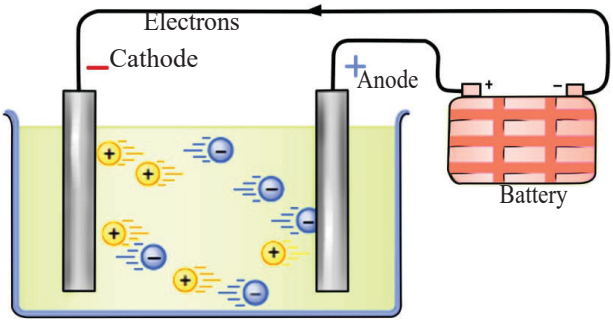


Fig. 5.7

Examine Figure 5.7 which represents the electrolytic process.

Ions are free to move in the liquid state or in an aqueous solution of an electrolyte. These ions are responsible for the electrical conductivity in the electrolyte.

Acids (HCl, H_2SO_4, HNO_3, CH_3COOH, ...) and alkalies (KOH, NaOH,...) contain free ions in their aqueous solutions. Salts (NaCl, KCl,.....) contain free ions in their aqueous solutions and molten state. For example, sodium chloride dissociates into Na^+ and Cl^- ions in the molten state or in the aqueous solution. Electrodes are conductors through which electricity enters or leaves the electrolyte. During electrolysis, the electrode connected to the positive terminal of the battery is called the anode and the electrode connected to the negative terminal is called the cathode. Reduction takes place at the cathode and oxidation at the anode.

In an electrolytic cell, oxidation takes place at the anode which is the positive electrode and reduction takes place at the cathode which is the negative electrode.

In a galvanic cell, the anode where oxidation takes place is represented as the negative electrode and the cathode, where reduction takes place is represented as the positive electrode.

Electrolysis of molten sodium chloride

In the solid state, sodium chloride does not conduct electricity. The reason is that, though the ions can vibrate about their fixed positions, they are not free to move. However, molten NaCl is a good conductor because its ions move freely. Examine Figure 5.8.

Two graphite electrodes are connected to a direct current (DC) source through wires. They are dipped in a vessel containing molten sodium chloride. When electricity is passed through them, the following observations can be made.

1. Chlorine gas (Cl_2), having a pale green colour, is liberated at the anode.
2. Silvery molten sodium metal (Na) is formed at the cathode.

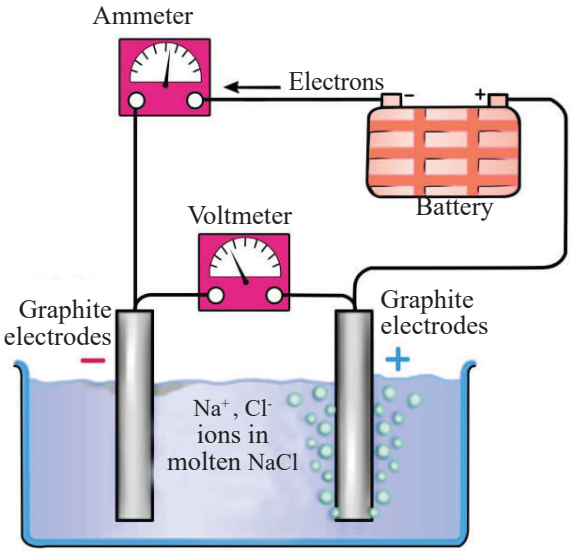


Fig. 5.8

What are the ions in molten sodium chloride?

.....

What is the chemical reaction taking place at the anode?

Oxidation / Reduction

Write the equation of the reaction. .....

What is the chemical reaction taking place at the cathode?

Oxidation / Reduction

Write the equation of the reaction.

.....

What are the products obtained as a result of the electrolysis of molten sodium chloride?

Electrolysis of aqueous sodium chloride solution

Examine Figure 5.9 representing the electrolysis of aqueous sodium chloride solution.

When a suitable voltage is applied across the electrodes of the cell, the following observations can be made.

1. H_2 gas is liberated at the cathode. If phenolphthalein indicator is present in the solution, the solution around the cathode (Figure 5.9) turns pink. That is, the solution becomes basic in nature.
2. Cl_2 gas is liberated at the anode.

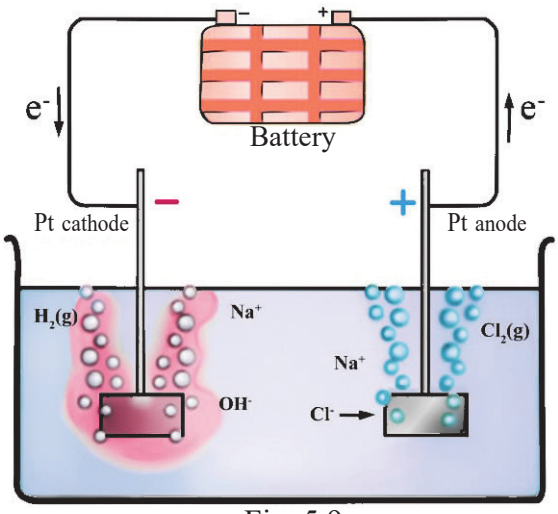


Fig. 5.9

Let us see how these observations can be explained.

At the anode, chloride ions are getting oxidised to Cl_2 gas.

Write the equation of the reaction.

.....

Based on the reactivity series, which has greater oxidising tendency- sodium or hydrogen?

If so which of the following is getting reduced at the cathode, Na^+ or H_2O?

At the cathode, H_2O molecules are reduced to produce H_2 gas and OH^- ions. Observe the equation for the reduction reaction.

$$2H_2O + 2e^- \rightarrow H_2 + 2OH^-$$

As a result of the processes taking place in the cell, gases such as H_2, Cl_2 and an aqueous solution of NaOH are formed. This aqueous solution can be evaporated to produce solid NaOH.

Electrodes	Chemical change	Product
Anode	$2Cl^- \rightarrow Cl_2 + 2e^-$	Chlorine gas
Cathode	$2H_2O + 2e^- \rightarrow H_2 + 2OH^-$	Hydrogen gas

Table 5.4

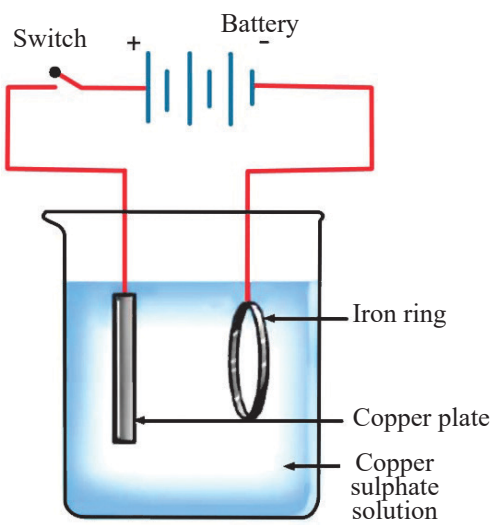


Fig. 5.10

Electroplating

Observe Figure 5.10.

Carry out this electrolysis with the help of teacher.

Record your observations.

Which metal is connected to the positive terminal of the battery? .....

Which metal is connected to the negative terminal of the battery? .....

Which solution is used as the electrolyte?

What are the ions present in the electrolyte?

Which electrode is connected to the positive terminal of the battery? Anode / Cathode

What is the reaction taking place at the anode?

Complete the equation of the oxidation reaction taking place at the copper plate.

$$Cu \rightarrow \ldots\ldots\ldots\ldots\ldots\ldots + 2e^-$$

Which electrode is connected to the negative terminal of the battery? Anode / Cathode

What is the reaction taking place at the cathode?

Which ions will be attracted to the cathode, the iron ring, from the solution?

Complete the equation of the reduction reaction taking place here.

$$Cu^{2+} + 2e^- \rightarrow \ldots\ldots\ldots\ldots$$

The reactions taking place at the anode and cathode:

Anode : $Cu \rightarrow Cu^{2+} + 2e^-$

Cathode : $Cu^{2+} + 2e^- \rightarrow Cu$

That is, at the anode, copper undergoes oxidation and enters into the solution as Cu^{2+} ions. At the same time, Cu^{2+} ions from the solution are reduced on the surface of the iron ring, which is the cathode, forming a thin coating of copper on it.

Electroplating is the process of coating a layer of one metal onto another metal through electrolysis.

The coating thus formed not only enhances the beauty of the metal but also prevents its corrosion. The object to be coated is connected to the negative terminal of the battery, after cleaning it. The metal which is used for plating is connected to the positive terminal of the battery. A salt solution of the metal to be plated is used as the electrolyte.

The metal to be coated	Electrolyte
Silver	Silver nitrate solution / sodium cyanide + silver cyanide solution
Gold	Sodium cyanide + gold cyanide solution

Table 5.5

Applications of electrolysis

1. Production of metals	:	Metals like sodium, potassium, calcium and aluminium are produced by the electrolysis of some of their compounds.
2. Production of nonmetals	:	Electrolysis is used in the industrial production of nonmetals like hydrogen, oxygen and chlorine.
3. Manufacture of compounds	:	Electrolysis is used in the production of sodium hydroxide and potassium hydroxide.
4. Electroplating	:	Gold plated jewellery, silver plated utensils, chromium plated iron objects etc are produced through electroplating.
5. Purification of metals	:	Electrolysis is useful in the purification of metals such as copper, gold etc.

Electrochemical cells as energy sources

A cell is a device that helps to convert the energy released in a chemical reaction directly into electricity. Cells perform two main functions.

1. As portable sources of electrical energy:

   Examples range from the button cells, used in electronic watches, to the lead acid cells used in vehicles.

2. As storage devices of electrical energy provided by an external source:

   Such cells can be used for powering electric vehicles, emergency power distribution and storing solar energy.

Different types of cells

There are two main types of cells.

Primary cells

In primary cells, the cell become dead, when the electrical energy produced by chemical reaction is used up. Such cells cannot be recharged and reused. Examples of primary cells are dry cells and button cells. Dry cells are commonly used in clocks. You may be aware that button cells are used in watches.

Secondary cells

A secondary cell can be recharged and used again. The most important type of secondary cell is the lead acid cell used in vehicles. The nickel-cadmium cell used in flashlights is also a secondary cell.

Lithium ion batteries are secondary cells that have become an integral part of our daily life. They are used as power sources for devices ranging from smartphones and laptops to electric vehicles. Lithium ion batteries are used in portable electronic devices due to their high energy density, long lifespan and low self discharge rate. Lithium ion batterries are also used as satellite batteries, a crucial component in the energy systems of spacecrafts.

Fuel cells

Fuel cells are galvanic cells designed to convert the chemical energy produced by the combustion of fuels such as hydrogen, methane and methanol into electrical energy. Such cells operate continuously as long as fuel is supplied. One of the advantages of fuel cells is that they produce electricity with high efficiency.

Dry cell

Button cell

Fig. 5.11

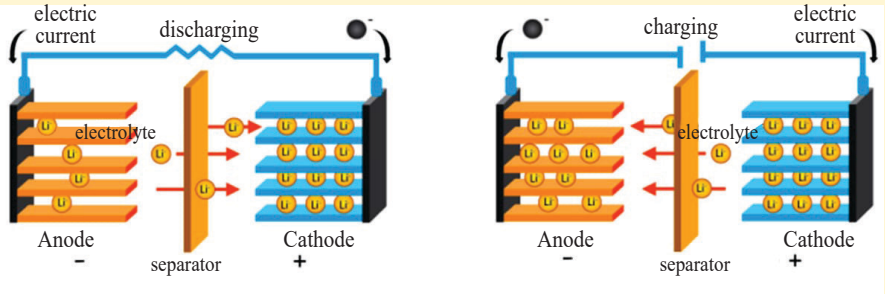
Lead acid cell

Nickel-cadmium cell

Fig. 5.12

Lithium ion batteries

The working principle of lithium ion batteries is the movement of lithium ions. The lithium ions move between the positive electrode (cathode) and the negative electrode (anode) in a liquid / gel polymer electrolyte.

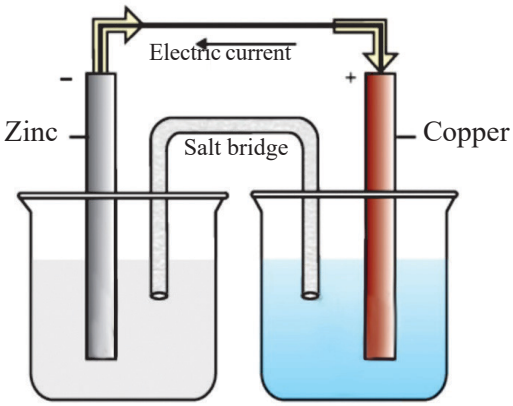


When a lithium ion battery is charged, lithium ions move from the cathode to the anode. This process allows the battery to store electrical energy. When the battery is discharging, lithium ions move from the anode to the cathode, which releases electrical energy to devices.

In 2019, Stanley Whittingham, John B. Goodenough, and Akira Yoshino won the Nobel Prize for Chemistry for their contributions to the development of lithium ion batteries.

Let us assess

1. Examine the diagram of a galvanic cell and find out whether the given statements are true or false.



   a) As the cell operates, the mass of the zinc rod, Zn(s) decreases.
   b) The copper electrode is the anode.
   c) Electrons flow from the zinc electrode to the copper electrode through the external circuit.
   d) During cell reaction, reduction takes place at the copper electrode.
   e) During cell reaction, the concentration of Cu^{2+} decreases.

2. The chemical equation of the reaction taking place in a galvanic cell is given.

$$Zn(s) + 2Ag^{+}(aq) \rightarrow Zn^{2+}(aq) + 2Ag(s)$$

   a) Represent diagrammatically the above galvanic cell.
   b) Which electrode has a negative charge?
   c) What is the reaction taking place at each electrode?

3. A galvanic cell is made up of the following half cells:
   - A magnesium electrode immersed in magnesium sulphate solution.
   - A nickel electrode immersed in nickel sulphate solution.

   The reactions taking place in the half cells are given.

$$Mg(s) \rightarrow Mg^{2+}(aq) + 2e^{-} \qquad Ni^{2+}(aq) + 2e^{-} \rightarrow Ni(s)$$

   Draw the cell diagram and label the anode, cathode and direction of flow of electrons.

4. When an aqueous sodium chloride solution is electrolysed, the reaction that takes place at the anode is:

   (i) $Na^{+}(aq) + e^{-} \rightarrow Na(s)$ (ii) $2H_2O(l) \rightarrow 4H^{+}(aq) + O_2(g) + 4e^{-}$

   (iii) $H^{+}(aq) + e^{-} \rightarrow \frac{1}{2} H_2(g)$ (iv) $Cl^{-}(aq) \rightarrow \frac{1}{2} Cl_2(g) + e^{-}$

5. Write the products obtained when the following solutions undergo electrolysis.

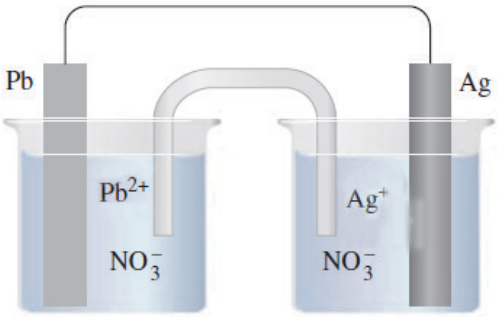
   (i) Aqueous solution of NaCl (ii) Aqueous solution of $CuSO_4$ (using copper electrodes)

6. Brass is an alloy of zinc and copper. When brass comes into contact with saline water, corrosion of metal takes place and zinc gradually dissolves in the solution, leaving copper behind. Explain why zinc dissolves in comparison to copper.
7. List the cells we use in our daily life and classify them into primary and secondary cells.
8. Define fuel cells and write its advantage.

Extended activities

1. The following are certain observations about metals A, B, C and D.
   (a) When a plate of metal A is placed in a solution containing B^{2+} ions, no reaction is observed.
   (b) When the plate of A is placed in a solution containing C^{+} ions, no change occurs.
   (c) When a plate of metal D is placed in a solution containing C^{+} ions, a black precipitate of C is formed on the surface of D and the presence of D^{2+} ions can be detected in the solution.
   (d) When a plate of B is placed in a solution of D^{2+} ions, D appears on the surface of B and B^{2+} ions appear in the solution.

   List A^{+}, B^{2+}, C^{+} and D^{2+} in the ascending order of their ability to attract electrons.
2. As shown in the galvanic cell of the given figure, place a silver electrode in a solution of silver nitrate and a lead electrode in a solution of lead nitrate. Connect the two electrodes using a copper wire. Also connect the two beakers using a salt bridge.
   Next, find the answers to the questions given below.



   (a) What is the anode? (b) What is the cathode? (c) Where does oxidation occur?
   (d) Where does reduction occur?
   (e) In which direction do electrons flow through the copper wire?
   (f) What will be the cell voltage?
   (g) Will the cell voltage vary if the two solutions are diluted alternately?